



ion — User Manual

Getting connected and using your ion server.

Note: This manual covers the **official TAK Server** variant of the ion server.

About your ion server

Each ion server is identified as `ion-xx`, where `xx` is the unit's serial number — for example, `ion-03` or `ion-04`. This identifier is both the WiFi network name (SSID) and the server's `.local` hostname.

The examples in this manual use `ion-03`. Substitute your unit's identifier throughout.

Connecting to the Server

There are three ways to reach the web UI. The WiFi access point works unconditionally; the other two require the server to be connected to a wired LAN or signed into a tailnet.

WiFi access point

1. Open your device's WiFi settings.
2. Join the network matching the unit's identifier — for example, `ion-03`.
3. Enter the WiFi password: `52235223`
4. Browse to `http://<hostname>.local` — for example, `http://ion-03.local` — or to the AP IP address listed on the info sheet shipped with the unit, for example `http://10.30.3.1`.

The AP IP address does not depend on mDNS and will always resolve. The web UI lists each interface IP and the server's Tailscale address, both of which are required for the two options below.

Wired LAN

Requires the server to be connected to an Ethernet network. From any device on that same LAN, browse to `http://<hostname>.local` — for example, `http://ion-03.local`.

Heads up: The `.local` address uses a network feature called *mDNS* (Bonjour). It works on most Macs, iPhones, iPads, Linux machines, and modern Windows and Android devices — but **it is not foolproof**. Some phones, some Windows configurations, some corporate/guest networks, and some VPNs will fail to resolve `.local` names.

If `.local` does not resolve, look up the server's LAN IP address over the WiFi access point:

1. Connect to the WiFi access point as described above.
2. Open the web UI and find the **Network** section. It lists every network interface on the server and the IP address currently assigned to each one.
3. Locate the Ethernet interface (typically named `enp1s0`, `eno1`, or `eth0`) and note its IP address — for example, `192.168.1.42`.
4. Disconnect from the WiFi access point and rejoin the LAN. Browse to `http://<that-ip-address>` — for example, `http://192.168.1.42`.

Tip: Write the LAN IP down (or save the bookmark). As long as the server stays on the same wired network it will typically keep the same address, so you only need to look it up once.

Tailscale

Requires the server to be signed into a tailnet and the client device to be signed into the same tailnet. Browse to the server's Tailscale IP address (`100.x.x.x`) or its MagicDNS name — for example, `ion-03.<tailnet>.ts.net`. Both are shown in the Tailscale panel of the Admin area.

`.local` names do not resolve over Tailscale; use the Tailscale IP address or the MagicDNS name.

QR Codes on the Dashboard

The dashboard's front page has a **Quick Connect** area with two QR codes you can scan with a phone camera.

Join WiFi

Scan this code with your phone's camera to join the server's WiFi network automatically — you don't have to type the network name or password.

Open Web UI

This code encodes the exact address the dashboard is currently being viewed at. Scan it from another device to open the same dashboard on that device — useful for handing the page off to someone nearby. The address is also shown as text directly beneath the code.

Certificate Downloads

WebAdmin Certificate

The **Download WebAdmin certificate** button downloads `webadmin.p12`. Install it in the web browser on the computer used to administer the TAK Server — it lets that browser log into the TAK Server web admin interface.

Intermediate CA Certificate

The **Download CA certificate** button downloads `caCert.p12`. Install it on each client device (ATAK / WinTAK), then set up the TAK server connection and log in.

Before a client device (EUD) can connect, an operator creates a user account on the server and then the EUD enrolls against that account using the intermediate CA certificate. The two subsections below cover each side.

Creating the user (server side)

1. From the TAK Server main screen, go to **Administrative** → **Add Users**.
2. In the upper left-hand corner, click **Add User**.
3. Create a username and password — be sure to read the password requirements shown on that field.
4. There is no need to add a group unless you want something other than the default `__ANON__` group.

Setting up the EUD for enrollment

1. Go to **Menu** → **Network Preferences** → **TAK Servers**.
2. In the upper right corner, tap the **3 dots** → **Add**.

3. **Name:** whatever you want.
4. **Address:** the server address — pull it from the web UI. Which address you use depends on how you want to reach the server: the AP IP, the Tailscale IP, or the LAN IP.
5. Check **Authentication** and enter the username and password from the server-side user creation.
6. Check **Enroll for Client Certificate**.
7. Check **Enroll with preconfigured trust**.
8. Check **Advanced Options** → **Import Trust Store** and select the certificate you downloaded via **Download CA cert** in the web UI. The password is `atakatak`.
9. Leave the **client cert** area empty.
10. Tap **OK** — the server should connect.

Admin Section

The dashboard has an **Admin** area gated by an operator PIN. Enter the PIN `52235223` and click **Unlock**.

Once unlocked, the Admin area lets you:

- Start, stop, and restart the server's services.
- View the **Mumble SuperUser** password (default `52235223`).
- Manage Tailscale — connect/disconnect, and log into or switch tailnets.

Click **Lock admin** to log back out.

Tailscale

Tailscale is a mesh VPN that gives the server a private address reachable from anywhere over the internet — handy for remote admin without exposing anything publicly.

From the Admin area:

- **Log into another tailnet** — starts a login and shows a sign-in link. Open it and sign in to your Tailscale account to connect the server.
- **Up / Down** — once logged in, connect or disconnect from the tailnet.
- **Switch** — if the server is signed into more than one tailnet, switch between them.

When connected, the panel shows the server's Tailscale address.

Services

The server runs several networked services. Reach them at one of the server's **IP addresses**:

- The **AP IP** — `10.30.xx.1`, where `xx` is the unit's serial number. For `ion-03` that is `10.30.3.1`. This always works while you are joined to the unit's WiFi.
- Its **LAN IP** — shown in the **Network** section of the dashboard.
- Its **Tailscale IP** (`100.x.x.x`) — shown in the Tailscale panel of the Admin area.

Use an IP, not `.local` : The `<hostname>.local` name may also work, but **do not use it when saving a server address inside an app**. Many mobile apps do not resolve `.local` names at all, and they never resolve over Tailscale — so a saved connection can fail with no obvious reason. Enter an IP address instead.

The examples below use `10.30.3.1` (the AP IP for `ion-03`). Substitute whichever of the server's addresses your device can reach.

MediaMTX — video streams

A place to send and view video streams. Publish once, then pull it back on any protocol — including **into ATAK**, so a feed becomes a live video layer for the whole team.

- Publish: RTSP `rtsp://<server-ip>:8554/<path>` — for example `rtsp://10.30.3.1:8554/cam1` — or RTMP `rtmp://<server-ip>:1935/<path>`.
- View: HLS `http://<server-ip>:8888` or WebRTC `http://<server-ip>:8889` — for example `http://10.30.3.1:8888`; add the RTSP URL as a video source in ATAK.

`<path>` is any name you pick (e.g. `cam1`); use the same path to publish and view.

Always use an IP address in the RTSP URL you add to ATAK — a `.local` name will often fail to resolve on the phone or tablet.

How to publish a stream

Publish a stream from **OpenTAK ICU**. In its settings:

- **Stream protocol**: RTSP
- **Stream address**: the server IP — the same one you entered for the server connection. Which address you use depends on how you are connected to the server.
- **Stream port**: default `8554`

- **Path:** whatever you want to call the stream. This example uses `mystream` .
- No username or password needed.
- **TCP:** enabled

Video preferences

- **Source:** this device's camera
- **Resolution:** whatever you want — `800x600` works well most of the time.
- **Bitrate:** `1000`
- **Adaptive bitrate:** on
- **FPS:** `15`
- **Codec:** H264

Audio settings

- **Audio enabled:** on
- **Bitrate:** `128`
- **Sample rate:** `44100`
- **Codec:** AAC
- **Stereo, echo canceller, and noise suppressor:** on

Adding the stream inside TAK Server

1. From the TAK Server front page, go to **Menu** → **Data** → **Video Feed Manager**.
2. Click **Add**.
3. **Alias:** whatever you want it labeled as in TAK Server.
4. **Protocol:** RTSP
5. **Address:** the server address
6. **Port:** `8554`
7. **Path:** from the EUD — in this case `mystream` .
8. Nothing else is required. Hit **Save**.

Mumble — voice

Voice server to point external apps at. Connect a Mumble client or the ATAK Mumble plugin to

`<server-ip>:64738` — for example `10.30.3.1:64738` .

Enter an IP address here, not `<hostname>.local` . The ATAK Mumble plugin and many mobile Mumble clients cannot resolve `.local` names, and the connection will fail without a clear error.

Reticulum

Background mesh networking stack (the `rnsd` daemon). Carries encrypted traffic over the mesh and is what MeshChatX runs on.

MeshChatX

Web app for the Reticulum mesh — LXMF messaging, network map, and page browsing. Open `http://<server-ip>:8000` — for example `http://10.30.3.1:8000` ; also linked from the dashboard's Services list.